



4G LTE single-mode modem by Samsung, operating in the first commercial 4G network by Tella (Courtesy: en.wikipedia.org)

signals travel over paired spectrum, while data communication works better on unpaired spectrum as people download more than upload.

The high bandwidth of a single-carrier radio delivers superb economies of scale vis-a-vis multi-radio legacy approaches. It also provides scope for considerably bigger capacity than 3G-3.5G technologies that are limited to 5MHz or smaller spectrum bandwidth. It must be noted that since LTE uses wider chunks of spectrum, data transfer on LTE-based 4G networks is nearly four times faster than on 3G. LTE with its spectrum flexibility has a good life ahead since it can be accommodated into many different swathes of spectrum.

Worldwide LTE scenario

Radio spectrum is the lifeblood of a mobile operator and has emerged as a very precious resource that is high in demand and low in supply. While regulators across the globe are looking to release new spectrum in order to enable broadband services at all levels. Spectrum in the 2.6GHz band is of interest across Europe. A big chunk, i.e., 140 megahertz of the spectrum (2x70 MHz), will be deployed for FDD services like LTE, while another 50 megahertz for the unpaired TDD band will most likely be used for WiMAX services. The 2.6GHz spectrum

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